Corda (2021) ‘The secret of planets’ perihelion between Newton and Einstein’ — math + physical-intuition check for the *Relativistic Newtonian Theory* foundations

Trey Morris with Claude Opus 4.7

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Author-review report: Corda (2021) “The secret of planets’ perihelion between Newton and Einstein” — math + physical-intuition check

To: Tepper Gill (DRQM I + *Relativistic Newtonian Theory* senior author) **From:** Trey Morris **Date:** 2026-05-30 **Re:** Issue #81, basis for Tepper’s Mercury perihelion derivation **Source paper under review:** Christian Corda, *Phys. Dark Univ.* **32** (2021) 100834, doi:10.1016/j.dark.2021.100834. PDF in repo at `Roadmapping/External_Papers/Secret_Corda.pdf`; marker-pdf conversion at `Roadmapping/Converted_Markdown/Secret_Corda/Secret_Corda.md`.

Cover note (for author review, not for journal submission)

You indicated that the perihelion derivation in your *Relativistic Newtonian Theory* paper takes Corda’s framework as its starting point. This report checks Corda’s math and physical intuition end-to-end so you can adjudicate whether the framework’s intended Mercury prediction is the Corda-style `pim/M + relativistic correction` template (the chain analysed in PR #83 giving `+37.79''/century`) or the proper apsidal-angle calculation of the modified force law (PR #83 doc 05 giving `-7.17''/century`).

Per CLAUDE.md author-review convention, the findings below are surfaced for your verdict — not announced as corrections. Every algebraic claim is verified symbolically via Wolfram MCP; every interpretive claim is flagged as substantive AI and tagged with author-review markers `[for Tepper:]`.

Per Crocco §5 compliance, this is substantive AI. The prompt-of-record is the user instruction sequence on issue #81 and the user’s 2026-05-30 ask “write the complete analysis in a markdown file for further review by Tepper.” No prose generation prompt beyond the user’s instructions was used.

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§1. Corda’s paper — structure (8 pages, 77 numbered equations)

§	Topic	Headline equation	Result
§1	Introduction, GR perihelion	Eq. (1): `Deltaphi_GR = 24pi^3 a^2 / [T_0^2 c^2 (1-e^2)] = 3pi r_g / [a(1-e^2)]`	`42.99''/c` for Mercury
§2	"Circular orbit" Newtonian back-reaction	Eq. (20): `Deltaphi = pim/M`	`44.66''/c` (Approach 1)
§3	Harmonic-oscillator / apsidal-angle	Eq. (31): `phi_0 = 2pi[3 + F'(r)r/F(r)]^(-1/2)`	same `44.66''/c` (Approach 2)
§4	Reduced-mass Kepler	Eq. (57): `T = T_0 (1+m/M)^(-1/2)`	same `44.66''/c` (Approach 3)
§5	Breakdown of `pim/M` for Venus / Earth	(numerical only)	`30x` and `51x` too large
§6	Gravitational time dilation	Eq. (67): `Deltaphi_F = (5pi/2) r_g/r_0`	`34.31''/c` (TD-only)
§7	Rotational (Langevin) time dilation	Eq. (72): `tau ~= (1 - 1/2 r^2omega^2/c^2) t`	(intermediate)
§8	Combined gravitational + rotational TD	Eq. (77): `Deltaphi_F = 3pi r_g/r_0`	`41.17''/c` (matches e=0 GR)
§9	Conclusion: "Newtonian + corrections matches GR"	—	—

The paper's two thesis claims (§9 verbatim):

(i) It is not correct that Newtonian theory cannot predict the anomalous rate of precession of the perihelion of planets' orbit. The real problem is instead that a pure Newtonian prediction is too large.

(ii) Perihelion's precession can be achieved with the same precision of general relativity by extending Newtonian gravity through the inclusion of gravitational and rotational time dilation effects.

The math + physical-intuition check below addresses each thesis separately.

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§2. Every load-bearing equation surfaced and verified

Each subsection below states Corda's equation verbatim (translated from the marker-pdf conversion), notes the Wolfram-MCP-verified algebra, and flags the physical-intuition status.

§2.1 Eqs. (3)–(5) — circular-orbit Newton baseline (no back-reaction)

`\frac{GMm}{r_0^2} = \frac{m v_0^2}{r_0}     \qquad (3)`

`v_0 = \sqrt{GM/r_0}                          \qquad (4)`

`T_0 = \frac{2\pi r_0}{v_0} = \frac{2\pi r_0^{3/2}}{\sqrt{GM}}  \qquad (5)`

Standard textbook (Newton + centripetal). [OK] Algebraically correct. Physical content: Mercury orbits a stationary Sun. Standard.

§2.2 Eq. (6)–(7) — Newtonian angular velocity and unperturbed angle

`\omega_0 = 2\pi / T_0                          \qquad (6)`

`\varphi_0 = \omega_0 T_0 = 2\pi                \qquad (7)`

Definitionally correct. [OK]

§2.3 Eqs. (8)–(13) — the substitution that drives Corda's §2 result

Corda's *core substitution*: replace `M` in Eq. (3) with `M + m` because the Sun-centred observer sees Mercury's reaction force on the Sun too.

`\frac{G(M+m)m}{r_0^2} = \frac{m v^2}{r_0}     \qquad (8)`

He justifies this by writing the Newton's-third-law pair (Eqs. 9–13):

`\vec F_G = \frac{GMm}{r_0^2}\hat u_r           \qquad (9)`

`M a_s\hat u_r = \frac{GMm}{r_0^2}\hat u_r \;\Rightarrow\; a_s = \frac{Gm}{r_0^2}  \qquad (10)`

`m a_m\hat u_r = -\frac{GMm}{r_0^2}\hat u_r \;\Rightarrow\; a_m = -\frac{GM}{r_0^2}  \qquad (11`

`a\hat u_r \equiv a_m - a_s = -\frac{G(M+m)}{r_0^2}\hat u_r  \qquad (12)`

`F\hat u_r = -\frac{G(M+m)m}{r_0^2}\hat u_r     \qquad (13)`

Algebra [OK] — this is the standard two-body relative-acceleration result. The same `G(M+m)` shows up in textbook reduced-mass treatments (Goldstein §3.1, MTW §25). What `(M+m)` captures is the kinematic fact that the *relative* acceleration of Mercury w.r.t. the Sun is bigger than `GM/r^2` by the factor `(M+m)/M = 1 + m/M`.

Physical intuition status [for Tepper:] what is being computed here is the **relative** acceleration in the centre-of-mass frame, not a "Sun-fixed" force on Mercury. The standard reduced-mass two-body decomposition replaces the two-body problem with a one-body problem for a reduced-mass particle `mu = Mm/(M+m)` orbiting a fixed centre at `M + m`. Corda's Eq. (12) is correct, but the interpretation in §2 (that this constitutes a "perturbation" to Mercury's force) muddles the standard story.

§2.4 Eqs. (14)–(17) — perturbed velocity and period

`v = \sqrt{G(M+m)/r_0}                          \qquad (14)`

`T = \frac{2\pi r_0}{v} = \frac{2\pi r_0^{3/2}}{\sqrt{G(M+m)}}  \qquad (15)`

`(M+m)^{-1/2} = M^{-1/2}\,(1 + m/M)^{-1/2}      \qquad (16)`

```
T = T_0\,(1 + m/M)^{-1/2}                          \qquad (17)
```

Wolfram MCP check:

```
ClearAll[GG, Msun, mMerc, r0, T, T0];
v   = Sqrt[GG (Msun + mMerc)/r0];
T   = 2 Pi r0 / v;
T0  = 2 Pi r0^(3/2) / Sqrt[GG Msun];
FullSimplify[T - T0 / Sqrt[1 + mMerc/Msun]]
(* Result: 0  [OK] *)
```

Algebra [OK]. Period decreases slightly because the effective gravitational constant in the one-body problem is `G(M+m)` not `GM`. This is a real kinematic effect on the **sidereal period**.

§2.5 Eqs. (18)–(20) — angular velocity and "perihelion advance"

```
\omega = \omega_0\,(1 + m/M)^{1/2}                 \qquad (18)
```

```
\varphi = \omega T_0 = 2\pi\,(1 + m/M)^{1/2} \simeq 2\pi\,(1 + m/(2M))  \qquad (19)
```

```
\boxed{\;\Delta\varphi = \varphi - \varphi_0 \simeq \pi m / M\;}  \qquad (20)
```

Wolfram MCP Taylor check:

```
Series[2 Pi Sqrt[1 + x] - 2 Pi, {x, 0, 2}]
(* Result: Pi x - Pi x^2/4 + O[x]^3  [OK] leading term is pi m/M *)
```

Algebra [OK]. Numerically for Mercury (`m/M = 1.66 x 10^-^7`): `Deltaphi = 5.21 x 10^-^7 rad/orbit x 415.2 orbits/century x (180.3600/pi) = 44.66''/century`. Wolfram-verified.

[X] Physical-intuition status [for Tepper:] here is the load-bearing critique. **What Eq. (20) computes is the angle Mercury sweeps in the unperturbed period `T_0` at the perturbed angular velocity `omega`.** That is not a perihelion advance. The perihelion advance is the angular rotation of the orbit's major axis between successive perihelia — a property of the shape of the orbit, not of the period or angular velocity.

For an inverse-square force the orbit is a closed ellipse (Bertrand's theorem), so the angle between successive perihelia is exactly `2pi`, **regardless of `T` and `omega`**. Changing `G` from `GM` to `G(M+m)` rescales the orbital speed and period but leaves the orbit a closed ellipse. The `pim/M` quantity Corda computes is a real kinematic effect, but it is not the perihelion-advance quantity. The §3 derivation makes this completely explicit — see §2.7 below.

§2.6 Eqs. (21)–(31) — Corda's own apsidal-angle formalism (§3 of paper)

Following Price & Rush (1979), Corda writes the radial equation, imposes angular-momentum conservation, Taylor-expands around `r_0`, and arrives at the **classical Bertrand-style apsidal angle**:

```
F_{c0}(r) = m\left(\ddot r - \omega_0^2 r\right)                \qquad (21)
```

```
J_0 = m r^2 \omega_0                                            \qquad (22)
```

```
F_{c0}(r) = m\left(\ddot r - \frac{J_0^2}{m^2 r^3}\right)       \qquad (23)
```

```
F_{c0}(r_0) = -\frac{J_0^2}{m r_0^3}                            \qquad (24)
```

After perturbing `x = r - r_0` and series-expanding:

```
\ddot x + m^{-1}\!\left[-\frac{3 F_{c0}(r_0)}{r_0} - F'_{c0}(r_0)\right] x = 0  \qquad (27)
```

```
T_0 = 2\pi\!\left(\frac{m}{-\,3 F_{c0}(r_0)/r_0 - F'_{c0}(r_0)}\right)^{1/2}  \qquad (28)
```

```
\frac{\varphi_0}{2} = \pi\!\left(\frac{m}{-3F_{c0}/r_0 - F'_{c0}}\right)^{1/2}\!\frac{J_0}{m r_
```

Using Eqs. (24) and (29):

```
\boxed{\;\varphi_0 = 2\pi\left[\,3 + \frac{F'_{c0}(r_0)\,r_0}{F_{c0}(r_0)}\,\right]^{-1/2}\;}
```

Wolfram MCP for `F = -k/r^2`:

```
ClearAll[Fc, r, r0, kk];
Fc[r_] := -kk/r^2;
ratio = (Fc'[r] r/Fc[r]) /. r -> r0;
phi0  = 2 Pi / Sqrt[3 + ratio];
{Simplify[ratio], Simplify[phi0]}
(* Result: {-2, 2 Pi}  [OK] For inverse-square: F'r/F = -2, phi_0 = 2pi, NO PRECESSION *)
```

Algebra [OK] — and this contradicts §2. Corda derives the apsidal-angle formula from scratch (Eq. 31), specialises to inverse-square in his own text (line right after Eq. 31: *"by setting F_{c0} = F_G in Eq. (31)... one finds phi_0 = 2pi, which is exactly Eq. (7)"*), and explicitly notes that this is the closed-orbit result. **For an inverse-square force, the apsidal angle is 2pi, the orbit is closed, and there is no precession.** Verified by Wolfram MCP (Bertrand's theorem at work).

§2.7 Eqs. (32)–(41) — Corda's "promotion" to the (1+m/M) force

To extend Eq. (31)'s apsidal-angle calculation to the reduced-mass case, Corda makes the substitution:

```
F_{c0}(r) \;\to\; F_c(r) = \left(1 + \frac{m}{M}\right) F_{c0}(r)  \qquad (32)
\omega_0 \to \omega                                                \qquad (33)
J_0 \to J = m r^2 \omega                                           \qquad (34)
```

This is meant to mirror the §2 substitution `M \to M+m` inside the central force law. He then writes:

```
F'_c(r_0) = \left(1 + \frac{m}{M}\right) F'_{c0}(r_0)             \qquad (36)
```

```
\boxed{\;\frac{F'_c(r_0)}{F_c(r_0)} \;=\; \frac{F'_{c0}(r_0)}{F_{c0}(r_0)}\;}  \qquad (37)
```

Wolfram MCP check:

```
ClearAll[Fc, Fc0, r, r0, mMerc, Msun];
Fc[r_] := (1 + mMerc/Msun) Fc0[r];
FullSimplify[Fc'[r0]/Fc[r0] - Fc0'[r0]/Fc0[r0]]
(* Result: 0  [OK] - the (1+m/M) factor cancels in F'/F *)
```

Algebra [OK], and [X] here is where Corda's apparatus contradicts his own §2 conclusion. The apsidal-angle formula Eq. (31) depends *only* on the ratio `F'/F`. The Eq. (37) Wolfram-verified identity says that constant rescaling of `F_{c0}` leaves `F'/F` unchanged. Therefore the apsidal angle for the rescaled force `F_c = (1+m/M) F_{c0}` is *the same* as for the unscaled force

`F_{c0}` — which is `2pi` (Eq. 31 with `F_{c0} = F_G = -k/r^2`). **The "perturbed" force still gives a closed orbit. No precession.**

Corda's subsequent Eqs. (38)–(41) derive `T = T_0(1+m/M)^(-1/2)` again from the period formula (28), then `omega = omega_0(1+m/M)^(1/2)`, then `phi = omega T_0 ~= 2pi(1 + m/(2M))` — the same `pim/M` as §2. **But this is exactly the §2 mistake repeated: a period change rephrased as a "perihelion advance."** Corda's own Eq. (31), applied to his own Eq. (32) force, *contradicts* the §3 conclusion.

§2.8 Eqs. (42)–(57) — reduced-mass Kepler derivation

Corda's §4 ("Third Kepler's law") works out the standard centre-of-mass + reduced-mass two-body decomposition:

$$\vec R = \frac{m\vec r_m + M\vec r_M}{M + m},\qquad \vec r = \vec r_m - \vec r_M \qquad (43)$$

$$J = \mu r^2 \omega = 2\mu\,\dot A \qquad (45)$$

$$T = 2\pi\!\sqrt{\frac{a^3}{G M_T}},\quad M_T = M + m \qquad (53)$$

$$\frac{a^3}{T^2} = \frac{G(M+m)}{4\pi^2} \qquad (54)$$

$$T = T_0 (1 + m/M)^{-1/2} \qquad (57)$$

Algebra [OK] — standard textbook treatment (Goldstein §3.7, MTW §25). Lands at the same `T = T_0(1+m/M)^(-1/2)` as Eqs. (17) and (39).

Physical-intuition status [for Tepper:] §4's derivation is impeccable as a derivation of the *period correction* due to the two-body kinematics. That's what the reduced-mass treatment gives. But Corda then writes Eq. (59) `phi = omegaT_0 ~= 2pi(1 + m/(2M))` and calls the excess a "perihelion advance" — same conflation as §2.7.

§2.9 Numerical summary of Corda's three approaches (§2/§3/§4 all give the same Deltaphi)

```
ClearAll[GG, Msun, mMerc, aMerc, Torb, orbitsPerCentury, radToArcsec, dphiNewton];
GG = 6.6743*^-11; Msun = 1.98892*^30; mMerc = 3.3011*^23;
aMerc = 5.7909*^10; Torb = 87.969*86400.;
orbitsPerCentury = 100 * 365.25 * 86400. / Torb;
radToArcsec = 180/Pi * 3600.;
dphiNewton = Pi mMerc / Msun;
Print["Deltaphi = pim/M per orbit   = ", N[dphiNewton], " rad"];
Print["Deltaphi = pim/M per century = ", N[dphiNewton * orbitsPerCentury * radToArcsec], " arcs
(* Result:                                                                       *)
(*   Deltaphi = pim/M per orbit   = 5.21424x10^-^7 rad                                    *)
(*   Deltaphi = pim/M per century = 44.6557 arcsec  <- Corda's reported 44.39''/c        *)
```

The numerical match to observed `43''/century` is correct arithmetic on a quantity that — per §2.6 and §2.7 above — is not** a perihelion advance.**

§2.10 §5 — Breakdown for Venus and Earth (the paper's self-refutation)

Corda himself shows the `pim/M` formula fails for non-Mercury planets:

```
ClearAll[mVenus, mEarth, TVenus, TEarth, orbitsVenus, orbitsEarth];
mVenus = 4.87*^24; TVenus = 224.701*86400.;
mEarth = 5.97*^24; TEarth = 365.256*86400.;
orbitsVenus = 100*365.25*86400./TVenus;
orbitsEarth = 100*365.25*86400./TEarth;
{"Venus",  N[Pi mVenus/Msun * orbitsVenus * radToArcsec], "vs observed 8.62''/c"}
{"Earth",  N[Pi mEarth/Msun * orbitsEarth * radToArcsec], "vs observed 3.83''/c"}
(* Results:                                                                    *)
(*   Venus  formula prediction: 257.91 ''/c   vs observed  8.62''/c  ->  30x too big *)
(*   Earth  formula prediction: 194.50 ''/c   vs observed  3.83''/c  ->  51x too big *)
```

Planet	`pim/M` (Corda's formula)	Observed residual	Discrepancy
Mercury	`44.66''/c`	`~= 43''/c`	[OK] 1.04x <- "agreement"
Venus	`257.91''/c`	`~= 8.62''/c`	[X] **30x too big**
Earth	`194.50''/c`	`~= 3.83''/c`	[X] **51x too big**

A formula that's right by 4% for one planet and wrong by factors of 30–50 for others is not a physics result — it is a numerical coincidence at one particular `m/M` value. Corda himself acknowledges the breakdown (§5) but rationalises it with n-body effects ("Venus has Mercury as an interior planet"), which does not explain why a strictly two-body formula would be valid for one two-body system and invalid for others. **The honest reading: `pim/M` is not the perihelion advance for any planet, including Mercury.**

§2.11 Eqs. (60)–(67) — gravitational time dilation

Corda's §6 applies Schwarzschild weak-field corrections. Standard relations:

```
t_g = \sqrt{g_{00}(r_0)}\, t_l                                     \qquad (60)
```

Isotropic-coordinate weak-field Schwarzschild line element:

```
ds^2 = \left(1 - \frac{2GM}{r c^2}\right)(c\,dt)^2
       - \left(1 + \frac{2GM}{r c^2}\right)(dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta\,d\phi^2)
                                                                   \qquad (61)
```

```
t_g = \sqrt{1 - r_g/r_0}\, t_l \;\approx\; \left(1 - \tfrac{1}{2}\tfrac{r_g}{r_0}\right) t_l
```

where `r_g = 2GM/c^2` is the Sun's gravitational radius (`~= 2.95 km`). Corda also asserts a corresponding radial-distance contraction: `r_0 -> r_0(1 - 1/2 r_g/r_0)` (Eq. 63), arguing from `r = ct`.

[partial] **Physical-intuition status [for Tepper:]** the time-dilation factor (Eq. 62) is standard GR weak-field. The radial-distance contraction in Eq. (63) is unusual — in isotropic coordinates the radial coordinate `r` of a point at Schwarzschild radius `R` is `r ~= R(1 - GM/(c^2R))` (a small isotropic-to-areal coordinate shift), so there *is* a `1/2 r_g/r_0` correction when comparing isotropic to areal radial coordinates, but treating this as "the CNO sees a different distance because `r = ct`"

is non-standard. Worth flagging but the final-result coefficient in Eq. (67) is unaffected to first order.

Substituting Eq. (63) into Eq. (17):

```
T_F = \frac{2\pi r_0^{3/2}\left(1 - \tfrac{1}{2}\tfrac{r_g}{r_0}\right)^{3/2}(1 + m/M)^{-1/2}}
```

Computing `omega_F = 2pi/T_F`:

```
\omega_F \simeq \omega\left(1 - \tfrac{1}{2}\tfrac{r_g}{r_0}\right)^{-3/2}\!\left(1 + \tfrac{1
```

```
\varphi_F \simeq 2\pi\left(1 + \tfrac{5}{4}\tfrac{r_g}{r_0}\right)   \qquad (66)
```

```
\boxed{\;\Delta\varphi_F \;\simeq\; \tfrac{5\pi}{2}\,\tfrac{r_g}{r_0}\;}  \qquad (67)
```

Wolfram MCP Taylor check:

```
Series[2 Pi (1 - x/2)^(-3/2) (1 + x/2), {x, 0, 1}]
(* Result: 2 Pi + (5 Pi/2) x + O[x]^2   [OK] leading is (5/2)pi r_g/r_0 *)
```

Algebra [OK] given the substitution in Eq. (63). This is roughly two-thirds of the GR answer.

§2.12 Eqs. (68)–(72) — rotational time dilation (Langevin)

Standard SR derivation of the Langevin line element for a rotating frame:

```
ds^2 = c^2 dt^2 - dr^2 - r^2 d\phi^2 - dz^2                              \qquad (68)
```

```
\phi = \phi' + \omega t'                                                  \qquad (69)
```

```
ds^2 = \left(1 - \tfrac{r^2\omega^2}{c^2}\right)c^2 dt^2 - 2\omega r^2 d\phi' dt' - dr^2 - r^2
```

```
d\tau = \sqrt{1 - r^2\omega^2/c^2}\,dt \;\simeq\; \left(1 - \tfrac{1}{2}\tfrac{r^2\omega^2}{c^
```

At `r = r_0`, `omega = omega_0`:

```
\tau \simeq \left(1 - \tfrac{1}{2}\tfrac{r_0^2\omega_0^2}{c^2}\right) t   \qquad (72)
```

Algebra [OK] — textbook special relativity. **Physical-intuition status:** Corda invokes Einstein's Equivalence Principle to argue that the centrifugal field in the rotating frame can be treated "as if" it were a gravitational field. This is the conceptual move that permits combining gravitational + rotational TD additively. [**for Tepper:**] the EEP application here is standard; the question is whether the Langevin correction, applied to a Keplerian orbital period, is the right way to recover the GR perihelion advance. The §8 result suggests yes (in the circular limit), and this matches the standard PPN result for the gravitational-redshift + Doppler combination.

§2.13 Eqs. (73)–(77) — combined correction

Using Eqs. (5)–(6) to express the rotational TD in terms of `r_g`:

```
\tau = \sqrt{1 - \tfrac{1}{2}\tfrac{r_g}{r_0}}\, t
     \simeq \left(1 - \tfrac{1}{4}\tfrac{r_g}{r_0}\right) t        \qquad (73)
```

The CNO replaces `T_F \to T_T = T_F (1 - \tfrac{1}{2} r_g/r_0)^{1/2}` and:

```
T_T = T_0 (1 + m/M)^{-1/2}\,(1 - \tfrac{1}{2} r_g/r_0)^2            \qquad (74)
```

```
\omega_F = 2\pi/T_T \simeq \omega\,(1 - \tfrac{1}{2} r_g/r_0)^{-2} \simeq \omega\,(1 + \tfrac{3
```

`\varphi_F = \omega_F T_0 (1 - m/(2M)) \simeq 2\pi\left(1 + \tfrac{3}{2}\,\tfrac{r_g}{r_0}\righ`

`\boxed{\;\Delta\varphi_F \;\simeq\; 3\pi\,\tfrac{r_g}{r_0}\;}  \qquad (77)`

> **OCR/marker-pdf note.** The conversion of Eq. (75) shows the last rotational factor as `(1 - r_g/(2r_0))^{+1/2}`, which would give a leading coefficient of `1` (not `3/2`). Reading instead as `(1 - r_g/(2r_0))^{-1/2}` (consistent with `omega = 2pi/T` and the `T_T = T_F (1 - r_g/(2r_0))^{1/2}` substitution) recovers the `3/2`. Either Eq. (75) in the paper has a typo or the converted markdown does. The final Eq. (77) is correct and matches the standard PPN gravitational-redshift + Doppler combination.

Wolfram MCP Taylor check (assuming -1/2):

```
Series[(1 - x/2)^(-3/2) (1 + x/2) (1 - x/2)^(-1/2), {x, 0, 1}]
(* Result: 1 + 3x/2 + O[x]^2   [OK] leading is (3/2) r_g/r_0 *)
```

Algebra [OK] (modulo the suspected typo in Eq. 75).

§2.14 Comparison of Corda's Eq. (77) to standard GR

Standard GR (Schwarzschild geodesic, MTW §40.5):

`\Delta\varphi_{\rm GR} = \frac{6\pi GM}{c^2 a (1-e^2)} = \frac{3\pi r_g}{a(1-e^2)}`

For Mercury (`e = 0.20563`, `1 - e^2 ~= 0.9577`): `Deltaphi_GR = 42.99''/century`.

For the **circular** limit (`e = 0`, `a = r_0`): `Deltaphi_GR = 3pi r_g/r_0`. **Corda's Eq. (77) reproduces the circular-orbit limit of GR.**

```
ClearAll[GG, Msun, aMerc, eMerc, cc, rg, orbitsPerCentury, radToArcsec];
GG = 6.6743*^-11; Msun = 1.98892*^30; aMerc = 5.7909*^10;
eMerc = 0.20563; cc = 299792458.; rg = 2 GG Msun / cc^2;
orbitsPerCentury = 100 * 365.25 * 86400. / (87.969 * 86400.);
radToArcsec = 180/Pi * 3600.;
{"Corda §8 (3pi r_g/a, circular)",  3 Pi rg/aMerc * orbitsPerCentury * radToArcsec,
 "GR Mercury  (3pi r_g/[a(1-e^2)])", 3 Pi rg/(aMerc (1 - eMerc^2)) * orbitsPerCentury * radToA
(* Results:                                                                      *)
(*   Corda §8 circular:  41.17''/c                                               *)
(*   GR Mercury (e=0.21): 42.99''/c                                              *)
```

Corda's §§6–8 derivation is missing the `1/(1-e^2)` eccentricity factor. For Mercury this is a `4.4%` shortfall (`-1.82''/century`). **[for Tepper:]** this discrepancy is significant for Mercury specifically because of its high eccentricity; for Venus and Earth (more circular) the discrepancy shrinks. A true derivation of the GR result needs the eccentricity from the genuine orbit (e.g.~via the apsidal-angle formula or the relativistic correction to the orbit equation).

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§3. The load-bearing critique — three findings

Finding 1 (algebra-driven). Bertrand's theorem rules out §2–§4

Corda's apsidal-angle formula (Eq. 31), applied to *any* inverse-square force — including the `(1+m/M)`-rescaled version of Eq. (32) — gives `phi_0 = 2pi` exactly: - `F = -k/r^2` => `F'(r) r / F(r) =`

-2 => `phi_0 = 2pi/sqrt(3 - 2) = 2pi`. [OK] - `F = (1+m/M)(-k/r^2)` => same `F'/F = -2/r` => same `phi_0 = 2pi`. [OK]

This is Bertrand's theorem: of all central potentials, *only* `1/r` (Kepler / Newton) and `r^2` (harmonic) give closed bound orbits universally. Inverse-square forces (Newton's gravity, Eq. 9) produce closed ellipses with no precession at any energy or angular momentum. Constant rescaling of an inverse-square force does not change this — it just changes the orbital period.

Therefore Eq. (20)'s `Deltaphi = pim/M` is not an apsidal precession. It is the angle by which Mercury's *position* differs from `2pi` when measured at time `T_0` (the unperturbed period) under the perturbed angular velocity `omega`. After the true perturbed period `T = T_0(1+m/M)^(-1/2)` has elapsed, Mercury has swept exactly `2pi` again and returned to perihelion. **No precession.**

Finding 2 (empirics-driven). The §5 self-refutation

Applying Corda's own formula `Deltaphi = pim/M x (orbits/century)` to Venus and Earth:

Planet	`m x orbits/c x radToArcsec / Msun x pi`	Observed	Discrepancy
Mercury	`44.66''/c`	`43''/c`	[OK]
Venus	`257.91''/c`	`8.62''/c`	[X] **`30x`**
Earth	`194.50''/c`	`3.83''/c`	[X] **`51x`**

Corda's response ("n-body effects, Mercury has no interior planets") doesn't work as a defence: - The formula `Deltaphi = pim/M` makes no reference to n-body effects. - The derivation in §§2/3/4 is strictly two-body throughout (no interior planets enter). - If the formula were the correct two-body result, it would apply to any two-body sub-problem and the n-body corrections would be smaller perturbations — not corrections of `factor 30-50`.

[for Tepper:] the strongest available evidence that `pim/M` is not the correct two-body perihelion advance is that the formula is empirically wrong for every planet other than Mercury. The Mercury "agreement" is coincidence at `m/M ~= 1.66 x 10^-^7` happening to give a number near `43''/c` once multiplied by Mercury's `415.2 orbits/century`. Venus and Earth have different `(m/M, orbits/c)` products and give wildly wrong predictions.

Finding 3 (formal-derivation-driven). §§6–8 reproduce only the e=0 GR limit

Corda's Eq. (77) `Deltaphi_F = 3pi r_g/r_0` is correct algebra (modulo the §2.13 typo) but recovers only the circular-orbit limit of the true GR result `3pi r_g/[a(1-e^2)]`. For Mercury (e = 0.21) this is a 4.4% shortfall (`-1.82''/century`).

[for Tepper:] §§6–8 is a real first-order GR derivation route; what it gets right is the "PPN sum of gravitational + Doppler time dilation" result, which is well-known in GR pedagogy. What it doesn't capture is the eccentricity dependence, which makes it numerically equivalent to GR only for nearly-circular orbits. For Mercury, the eccentricity correction matters; for Venus and Earth (smaller `e`), `3pi r_g/a` is much closer to the full GR answer.

The two §§6–8 results work in their domain. The trouble is purely the combination with §§2–4. Corda writes (§9) that "Newtonian + corrections matches GR." The honest reading: §§6–8 *alone* matches GR (in the circular limit); §§2–4 contribute a fictitious `pim/M` that doesn't correspond to a real precession; the two together don't add up to GR — the `pim/M` is just spurious.

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§4. Implication for Tepper's *Relativistic Newtonian Theory*

Gill's *Relativistic Newtonian Theory* extends Corda's framework by adding the dual-theory `(1 + V/(mc^2))` factor to the force law:

$$a_m = -\frac{GM}{r^2}\left(1 - \frac{GM}{c^2 r}\right)\hat u_r \qquad (\text{Eq. h4})$$

(transcribed from PR #83's `Equation_Verification/Dual_Newtonian_Theory.md`; all algebra Wolfram-MCP verified).

§4.1 What PR #83 computed (Corda-style template)

PR #83 `03_numerical_predictions.md` followed Corda's heuristic:

$$\Delta\varphi_d = 2\pi\!\left[\left(1 + \tfrac{m}{M}\right) \left(1 - \tfrac{G(M^2 + m^2)}{(M+m) c^2 r_0}\right)\right]^{1/2} - 2\pi$$

with first-order Taylor expansion:

$$\Delta\varphi_{d_1} \;\simeq\; 2\pi\!\left[\frac{m}{2M} - \frac{GM(1+m/M)}{2 c^2 r_0}\right]$$

This decomposes as:

Contribution	Value	Source
Reduced-mass `pim/M`	`+44.66''/c`	Corda §§2–4 (**not a real precession**)
Framework relativistic correction `-piGM/(c^2 r_0)`	`-6.87''/c`	wrong sign vs. GR's `+42.99''/c`
Net `Deltaphi_d`	`+37.79''/c`	sum, vs. observed `~= 43''/c`

The 12% shortfall (`+37.79''` vs `+43''`) was interpreted in PR #83 as a near-miss. **Under the present analysis, this is a coincidence on top of a coincidence**: `pim/M` is fictitious, and the relativistic correction has the wrong sign.

§4.2 What PR #83 doc 05 computed (apsidal-angle, the genuine answer)

`Mercury_Perihelion/05_nbody_orbital_mechanics.md` applied Corda's *own* apsidal-angle formula (Eq. 31) to Gill's modified force law:

$$F(r) = -\frac{GMm}{r^2}\left(1 - \frac{GM}{c^2 r}\right)$$

The apsidal-angle formula gives a precession per orbit that is:

$$\boxed{\;\Delta\varphi_{\rm framework}^{\rm genuine} \;=\; -\frac{1}{6}\,\Delta\varphi_{\rm GR} \;=\; -7.17\,\text{''/century}\;}$$

Opposite sign, one-sixth magnitude of GR exactly. This is the famous "1/6 factor" of a force-law-only modification of Newtonian gravity that lacks GR's spatial-metric curvature. (See e.g. Misner-Thorne-Wheeler *Gravitation* §40.6 for the analogous PPN parameter analysis.)

The N-body refinement cannot close this gap: planetary perturbations contribute framework relativistic corrections at `~10^{-4}-10^{-5}` of the Sun–Mercury term (since `GM/(c^2r)` scales with the attracting mass and the other planets are light). The `-7.17''/century` is the framework's structural prediction for the eccentric two-body case, computed via the same apparatus (apsidal angle) Corda himself uses.

§4.3 The recommendation [for Tepper]

The Corda-style `pim/M + relativistic correction` template that *Relativistic Newtonian Theory* inherits should be replaced by the apsidal-angle calculation as the framework's genuine perihelion prediction. The relevant numbers:

Calculation	Result	Status
Corda-style heuristic (PR #83 doc 03)	`+37.79''/century`	numerical near-miss; structurally fictitious
Apsidal-angle (PR #83 doc 05)	`-7.17''/century`	genuine precession of modified force law
Standard GR	`+42.99''/century`	observed `~43''/c`

If the framework's intended Mercury prediction is the apsidal-angle result, the verdict is: **wrong sign, `1/6` magnitude**. This is the structural signature of any modified-Newtonian gravity that adds a `(1 + V/(mc^2))`-style factor to the force without modifying the spatial metric. The 12% near-miss in PR #83 doc 03 is an artefact of taking Corda's `pim/M` heuristic as the baseline.

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§5. Specific questions for Tepper

Six questions for your verdict, drawn from the analysis above:

1. **Bertrand's theorem applicability.** Do you agree that Eq. (31) applied to an inverse-square force (including the `(1+m/M)`-rescaled version of Eq. 32) gives `phi_0 = 2pi`, i.e.~no apsidal precession? This is the load-bearing critique of §§2–4. If you read Bertrand differently or identify a step in Eq. (31)'s derivation that doesn't apply, please flag.

2. **Period vs. precession.** Do you agree that the `pim/M` in Corda's Eqs. (19)–(20) is computing the angle Mercury sweeps in the *unperturbed* period `T_0` at the *perturbed* angular velocity `omega` — not the rotation of the orbit's major axis between successive perihelia? This is the conceptual distinction that determines whether *Relativistic Newtonian Theory* inherits a fictitious baseline.

3. **§5 self-refutation.** The same `pim/M` formula gives `Venus 258''/c` and `Earth 195''/c` vs. observed `8.6''/c` and `3.8''/c`. Is Corda's "n-body effects" rationalisation sufficient, or does the empirical failure mean the formula is not a perihelion advance? (PR #83's reading is the latter; we'd like your reading recorded.)

4. **§§6–8 vs. eccentricity.** Corda's `3pi r_g/r_0` reproduces the e=0 GR limit and misses `1/(1-e^2)`. For Mercury this is a 4.4% shortfall. Do you read §§6–8 as a valid pedagogical derivation of the circular-orbit GR result (PPN sum of gravitational + Doppler TD), or as a complete GR derivation? The distinction matters for whether the eccentricity correction needs to be added to the framework's prediction.

5. **The apsidal-angle calculation for Gill's force law.** PR #83 doc 05's apsidal-angle result `-7.17''/century` (`-1/6` of GR exactly) for the framework's force law `a = -(GM/r^2)(1 - GM/(c^2r)).ê_r` is the proper answer to "what does Gill's modified Newton predict for Mercury?". Do you read this as the framework's intended prediction, or is the Corda-style heuristic (`+37.79''/century`) intended?

6. **The `1/6` structural signature.** PR #83 doc 05 identifies `-1/6` of GR as the classic signature of a force-law-only modification of Newton without spatial-metric curvature. Does this match your understanding of what *Relativistic Newtonian Theory* delivers, or does the framework include a spatial-metric component that PR #83 missed?

The answers to these six adjudicate whether *Relativistic Newtonian Theory*'s Mercury prediction is `+37.79''/century` (Corda template), `-7.17''/century` (apsidal-angle), or something else.

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§6. Honest framing — what this report does and does not claim

What this report claims

- Corda's algebra is mostly correct (specifically: Eqs. 17, 20, 31, 37, 67, 77 are all Wolfram-MCP verified).
- Corda's §§2–4 result `Deltaphi = pim/M` is *not* a perihelion advance — it is a period-and-angular-velocity rescaling that does not change the apsidal angle of a closed inverse-square orbit. This is verified via Corda's own Eq. (31).
- The §5 numerical breakdown for Venus and Earth is empirical evidence that the formula is not a real perihelion advance.
- §§6–8 reproduce the GR result in the circular-orbit limit only.
- *Relativistic Newtonian Theory*'s `+37.79''/century` prediction inherits the Corda framing's fictitious `pim/M` baseline plus a wrong-sign relativistic correction.
- The framework's *genuine* perihelion prediction, computed via the apsidal-angle formula on the modified force law, is `-7.17''/century` (`-1/6` of GR).

What this report does not claim

- That Corda's paper is not a valid published work. It is a peer-reviewed paper in *Physics of the Dark Universe* (a respectable Elsevier journal). The findings here are about the *interpretation* of correctly-derived algebra.
- That *Relativistic Newtonian Theory* as a whole is wrong. The Mercury perihelion is one test; other tests (the four candidate curved-spacetime extensions tracked in the GPS author report's Q1) remain open.
- That the framework cannot deliver a Mercury-matching prediction. The framework may have alternative extensions to gravity beyond the one analysed in *Relativistic Newtonian Theory*; this report only addresses the specific extension that paper proposes.

- That the apsidal-angle critique is novel. It is straightforward textbook orbital mechanics. The novelty here is applying it explicitly to Corda's framework and to *Relativistic Newtonian Theory* as an inheritance check.

What the report is asking from you

Six adjudications (§5 above). Your verdict on these determines what the campaign's Mercury verdict should read; the campaign's current PR #83 documents both the Corda-style heuristic and the apsidal-angle critique and is held pending your reading.

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§7. Cross-references

- **This document**: `Roadmapping/Author_Reports/2026-05_corda_perihelion_review_for_gill.md`
- **Corda PDF**: `Roadmapping/External_Papers/Secret_Corda.pdf`
- **Corda markdown** (marker-pdf): `Roadmapping/Converted_Markdown/Secret_Corda/Secret_Corda.md`
- **Concise equation-verification doc**: `Roadmapping/Equation_Verification/Secret_Corda_Perihelion.`
- **Gill's *Relativistic Newtonian Theory* analysis (PR #83)**:
  - PR: https://github.com/temoTxt/PyPhysics/pull/83
  - `Roadmapping/Equation_Verification/Dual_Newtonian_Theory.md`
  - `Roadmapping/Mercury_Perihelion/README.md`
  - `Roadmapping/Mercury_Perihelion/05_nbody_orbital_mechanics.md` (apsidal-angle calculation; the `-1/6` of GR result)
- **GPS author report Q1 (curved-spacetime extension question)**: `Roadmapping/Author_Reports/2026-`
- **Issue #81**: https://github.com/temoTxt/PyPhysics/issues/81

§8. Human-acceptance section (Crocco §5 substantive AI)